

Passive Survivability: Designing for Tomorrow's Disasters

Given likely future events, homes should be designed to maintain livable conditions even when power supplies are interrupted.

by Alex Wilson

One of the few certainties coming out of the back-to-back natural disasters of Hurricanes Katrina and Rita is that these will not be the last storms to cause significant damage. There will be other storms that result in widespread destruction and power outages, especially given that more severe storms are predicted, as a result of global climate change. There is also a risk of disasters caused by human beings—terrorist actions—that could result in similar power loss. On top of this, as the demand for oil and natural gas begins to exceed the available supply, there is a risk that fuel supplies will be interrupted.

A new design approach has been advanced by the editors of *Environmental Building News* and participants in a series of Gulf Coast reconstruction charrettes sponsored by the U.S. Green Building Council as a way of dealing with these concerns. The name of this approach is passive survivability. Passive survivability involves designing homes and other buildings to maintain livable conditions in the event of extended loss of power, heat, or water. It is a strategy achieved by many of the design features that *Home Energy* magazine and other proponents of energy-efficient, green home building have advanced for decades—but the driver is a little different. The primary motivation of passive survivability is not to protect the environment or to save money; rather, it is to provide for the safety and well-being of people. The goal is to avoid problems like these:

- the weeklong Chicago heat wave in 1995 that killed more than 700 people;

- the January 1998 ice storm in Eastern Canada that forced 600,000 people from their homes; and

- the blistering temperatures in the New Orleans Superdome when it was being used as an emergency shelter following Hurricane Katrina.

Achieving Passive Survivability

The usual approach that homeowners take to protect themselves from power outages is to purchase and install emergency backup generators and enough fuel to maintain those generators during an extended outage. This is a reasonable strategy in some situations and should be considered as one option. However, generators are expensive to buy and maintain, and unless they are very large, they usually cannot provide cooling. The storage of needed fuel (especially liquid fuel) poses environmental and safety risks, and some fuels go bad after extended storage. The more volatile components of gasoline evaporate and the additives degrade; diesel is susceptible to bacterial growth. Propane is fairly stable and can be used as a backup for natural gas.

Proponents of passive survivability argue that it's better to rely on passive measures to protect a home and its occupants from energy and water supply interruptions. Here are some of the important strategies that they recommend.

High-performance envelope.

The most important strategy for achieving passive survivability is to provide a high-performance building envelope.

For cooler climates, this means high levels of insulation (minimum R-25 walls and R-40 ceiling/roof); very high-performance glazings (unit U-factor of 0.25 or lower, which usually requires low-e coatings, low-conductivity gas fill, and either triple glazing or double glazing with an interior film); full slab or foundation wall insulation; and airtight construction (maximum 5 ACH₅₀, with mechanical ventilation).

Cooling load avoidance. The easiest way to keep a house from getting too hot if air conditioning equipment cannot operate is to incorporate cooling load avoidance strategies. These include carefully orienting the house to minimize large areas of east- and west-facing glass; using low-solar-heat-gain-coefficient (SHGC) glass for east and west windows and skylights (values of 0.5 or lower); providing overhangs, awnings, or vegetative shading for south, east, and west windows; providing an Energy Star-rated reflective roof; and using radiant barriers or radiant-barrier roof sheathing in unheated attics.

Natural ventilation. Natural ventilation provides a way to exhaust hot air from a house without the use of fans. New houses can be configured to provide for natural air flow through the careful placement of windows, general building geometry, and the use of cupolas and other features to exhaust hot air near the peak of the roof.

Passive-solar heating. Most heating equipment will not operate in a power outage, because fans, pumps, and (often) controls require electricity—even if there is plenty of gas or oil. Depending on where a house is located, how it is

built, and on the season, not being able to turn on the heat can make the difference between maintaining a livable house and working plumbing, or having to move out. A well-designed passive-solar house with no backup heat can, in most climates, maintain a nighttime winter temperature of 55°F or higher—a level that will keep the occupants comfortable, if they have enough blankets and clothing.

Key passive-solar heating strategies include paying careful attention to the siting and orientation of the house, providing adequate south-facing glazing, using high-SHGC glass (minimum 0.7) in the south-facing windows, using high-mass materials within the conditioned envelope, and designing the overall building geometry and layout to facilitate good circulation of sun-warmed air. Using advanced software tools, such as *Energy-10*, will help to achieve success with passive-solar design.

Daylighting. While it is more critical to provide natural daylighting in schools and other public-use buildings that could be used as emergency shelters, natural daylighting is also important in homes. Strategies include careful placement of windows; use of south-facing clerestory windows that can bring daylight (as well as wintertime heat) deep into a house; installation of high-performance conventional skylights; installation of tubular skylights; and use of light-colored, reflective ceiling and wall finishes. For energy savings during times when there is electricity, particularly in schools and commercial buildings, electric lighting controls can be used that automatically dim or turn off lights when the daylighting is adequate.

Water storage. In many areas, an extended power outage would also mean loss of water. Rainwater catchment systems can provide potable water in an emergency. Even if rainwater is usually used only for outdoor irrigation, that same water can be used indoors for toilet flushing, washing, and—after filtration or treatment—for drinking and cooking. A century ago, most houses in New Orleans had rainwater collection and storage systems, and they are still common in parts of Appalachia.

Solar water heating. Either passive-solar water-heating systems or

active-solar water-heating systems powered by PV panels can provide hot (or at least warm) water during extended power or heating fuel outages. The fact that solar water heaters can operate without grid electricity is something to think about when one chooses a water heater.

PV power. Producing electricity from sunlight using PV panels is perhaps the ultimate in self-reliance. To function during an extended power outage when the sun isn't shining, however, a PV system has to include electricity storage using deep-cycle batteries. Many PV systems today are connected to the grid. This eliminates the need to store a battery on site, and it enables the occupants to benefit from net metering laws that allow them to run the electric utility's meter backward when the PV system is generating more electricity than the house is consuming. (Thirty-nine states and the District of Columbia have net metering laws that allow homeowners to sell electricity back to their electric utility at the retail price of that electricity—up to the amount of electricity they consume.) As a safety measure, grid-connected PV systems shut down the grid connection when the electric grid goes down, so that the PV system is not sending electricity into utility wires when linemen may be working on repairs. Thus, systems without batteries cannot operate when the grid is down. Special controls are needed to enable the occupants to use electricity during a power outage; for homeowners and builders who are striving to ensure passive survivability, I recommend that these controls be installed.

Redesigned heating equipment. As noted above, most heating equipment relies on electricity to operate fans, pumps, and controls. To enable this equipment to operate during an extended power outage, it could be redesigned to run on DC power, which could be generated by an integrated PV system. This PV-powered heating equipment could not operate at night (unless battery storage was provided), but in a well-insulated home, daytime operation should be adequate to maintain livable condi-

tions. Gas- or oil-fired heating equipment would still require a gas or oil supply to operate, so it would go out if the fuel supply was interrupted—but power outages are more common than fuel supply interruptions, so this approach affords a significant level of protection.

The Bottom Line

Providing for passive survivability in our buildings may be one of the most important design criteria today. Most passive-survivability strategies are identical to the strategies that proponents of energy-efficient or green building have been advocating for decades—only the motivation is different. Most proponents of energy efficiency and green building have to date been motivated by environmental concerns, or by the need to save costs. Passive survivability opens the door to a new type of advocate: those concerned about storm resilience and safety in the event of natural and human-caused disasters.

Code officials and code-writing organizations should take note. Building codes are supposed to protect us by ensuring that buildings are safe. Perhaps it's time to incorporate measures of passive survivability into those codes. 

Alex Wilson is the founder and president of BuildingGreen, Incorporated, in Brattleboro, Vermont, and executive editor of Environmental Building News. He is the editor of the recently released New Orleans Principles.

For more information:

Visit www.buildinggreen.com to read *Environmental Building News*.

New Orleans Principles can be downloaded at www.green_reconstruction.buildinggreen.com.

To download *Energy-10* software, go to the Sustainable Buildings Industry Council Web site, at www.SBICouncil.org.